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OSS Practice Week-10

1.Identify the command used to search for all failed login attempts (HTTP 401 errors) within a web server access log located in the home directory.

```
[analyst@secOps ~]$ grep " 401 " /home/analyst/lab.support.files/apache_in_epoch
.log
198.51.100.213 - - [1219071600] "GET /twiki/bin/edit/Main/Double_bounce_sender?t
opicparent=Main.ConfigurationVariables HTTP/1.1" 401 12846
[analyst@secOps ~]$
```

The grep " 401 " /home/analyst/lab.support.files/apache_in_epoch.log command searches the web server access log for entries with HTTP status code **401**, which indicate failed login attempts.

2.A suspicious hidden file is found in the /tmp directory; which command and flags should you use to view its detailed metadata, including the exact time of last modification?

```
[analyst@secOps ~]$ stat /tmp/.Test-unix
  File: /tmp/.Test-unix
  Size: 4096            Blocks: 8          IO Block: 4096   directory
Device: 802h/2050d      Inode: 785655      Links: 2
Access: (1777/drwxrwxrwt)  Uid: (    0/    root)   Gid: (    0/    root)
Access: 2026-02-10 14:12:01.215999899 +0000
Modify: 2026-02-10 14:12:01.215999899 +0000
Change: 2026-02-10 14:12:01.215999899 +0000
 Birth: 2026-02-10 14:12:01.215999899 +0000
```

The stat command displays detailed metadata for the hidden file, including the exact last modification timestamp.

3.You suspect a hidden background process is running without a terminal; what command would you use to list all running processes and filter for those that have no associated TTY?

```
[analyst@secOps ~]$ ps aux | awk '$7=="?"'
root           1  0.1  0.3 167140 12528 ?        Ss   09:11   0:01 /sbin/init sp
lash
root           2  0.0  0.0      0     0 ?        S    09:11   0:00 [kthreadd]
root           3  0.0  0.0      0     0 ?        I<   09:11   0:00 [rcu_gp]
root           4  0.0  0.0      0     0 ?        I<   09:11   0:00 [rcu_par_gp]
root           5  0.0  0.0      0     0 ?        I<   09:11   0:00 [slub_flushwq
]
root           6  0.0  0.0      0     0 ?        I<   09:11   0:00 [netns]
root           8  0.0  0.0      0     0 ?        I<   09:11   0:00 [kworker/0:0H
-events_highpri]
root           9  0.1  0.0      0     0 ?        I    09:11   0:00 [kworker/u8:0
-ext4-rsv-conversion]
root          10  0.0  0.0      0     0 ?        I<   09:11   0:00 [mm_percpu_wq
]
root          11  0.0  0.0      0     0 ?        S    09:11   0:00 [rcu_tasks_ru
de_]
root          12  0.0  0.0      0     0 ?        S    09:11   0:00 [rcu_tasks_tr
```

This command lists all running processes and filters those with no associated terminal (TTY ?), which may indicate background or potentially suspicious processes.

4.How can you identify which specific process ID (PID) is currently maintaining an active network connection to an unknown external IP address?

```
[analyst@secOps ~]$ lsof -i -nP
COMMAND  PID    USER   FD   TYPE DEVICE SIZE/OFF NODE NAME
firefox 2765 analyst   49u  IPv4  64936      0t0  TCP 10.0.2.15:48996->151.101.1
.91:443 (ESTABLISHED)
firefox 2765 analyst   75u  IPv4  64955      0t0  TCP 10.0.2.15:36164->146.75.12
1.91:443 (ESTABLISHED)
firefox 2765 analyst   87u  IPv4  54767      0t0  TCP 10.0.2.15:47362->146.75.12
1.91:443 (ESTABLISHED)
firefox 2765 analyst  121u  IPv4  35996      0t0  TCP 10.0.2.15:47302->34.107.24
3.93:443 (ESTABLISHED)
```

The lsof -i -nP command lists all processes with active network connections and displays their corresponding PID, allowing identification of the process communicating with an external IP address.

5.To ensure a script runs every time the system boots, an attacker might modify a specific configuration; where should you look for unauthorized scheduled tasks for a specific user?

```
[analyst@secOps ~]$ crontab -l
no crontab for analyst
[analyst@secOps ~]$ sudo crontab -l
[sudo] password for analyst:
no crontab for root
```

Unauthorized scheduled tasks for a specific user are stored in the user's crontab (typically under /var/spool/cron/), which can be viewed using crontab -l.

6.What is the correct command to modify a user's account settings so that their password is required to be changed every 60 days?

```
[analyst@secOps ~]$ sudo chage -M 60 analyst
[analyst@secOps ~]$
```

The chage -M 60 command sets the maximum password age to 60 days, requiring the user to change their password after that period.

7.After a suspected system breach, you need to verify if the /bin/ls binary has been tampered with; describe the process and the specific tool used to check its integrity.

```
[analyst@secOps ~]$ sha256sum /bin/ls
8696974df4fc39af88ee23e307139afc533064f976da82172de823c3ad66f444  /bin/ls
[analyst@secOps ~]$
```

Generated hash:
8696974df4fc39af88ee23e307139afc533064f976da82172de823c3ad66f444
This hash should then be compared with a known trusted reference hash to determine whether the binary has been tampered with.

8.Use the find command to locate all files within a specific directory that have "world-writable" (777) permissions and explain why this is a security risk.

```
[analyst@secOps ~]$ find /home/analyst -type f -perm 777
[analyst@secOps ~]$
```

This command searches the specified directory (e.g., /home/analyst) for files that have world-writable (777) permissions, which are dangerous because any user can modify them.

9.You need to block all outgoing traffic to a specific malicious IP address while still allowing web browsing; provide the necessary iptables rule to achieve this.

```
[analyst@secOps ~]$ sudo iptables -A OUTPUT -d 10.0.2.4 -j DROP
[analyst@secOps ~]$
```

These iptables rules block all outgoing traffic to the specified malicious IP address while allowing normal web browsing through ports 80 and 443.

To still allow web browsing (HTTP/HTTPS):

```
[analyst@secOps ~]$ sudo iptables -A OUTPUT -p tcp --dport 80 -j ACCEPT
[analyst@secOps ~]$ sudo iptables -A OUTPUT -p tcp --dport 443 -j ACCEPT
[analyst@secOps ~]$
```

10.During a vulnerability assessment, you need to determine the exact version of the running SSH service; which nmapflag is required to perform service version detection on the local host?

```
[analyst@secOps ~]$ nmap -sV localhost
Starting Nmap 7.80 ( https://nmap.org ) at 2026-02-10 09:39 UTC
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000061s latency).
Not shown: 997 closed ports
PORT   STATE SERVICE VERSION
21/tcp open  ftp     vsftpd 2.0.8 or later
22/tcp open  ssh     OpenSSH 8.9p1 Ubuntu 3ubuntu0.1 (Ubuntu Linux; protocol 2.0
)
23/tcp open  telnet  Linux telnetd
Service Info: Host: Welcome; OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap
.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 11.48 seconds
```

The -sV flag enables service version detection, allowing Nmap to identify the exact version of services such as SSH running on the target host.